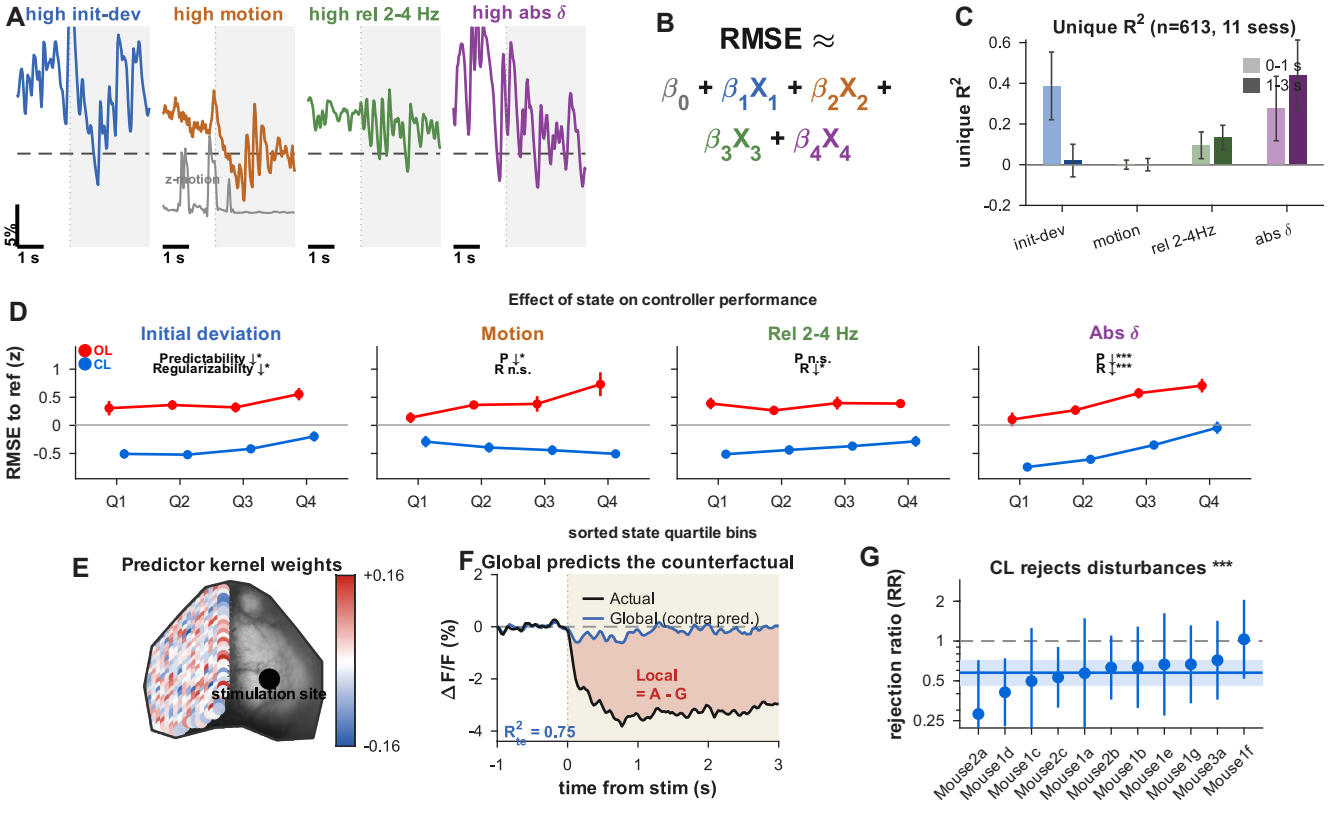

A
high init-dev
high motion
high rel 2-4 Hz
high abs δ
z-motion
5%
1 s
1 s
1 s
1 s
B
RMSE ≈
β0 + β1X1 + β2X2 +
β3X3 + β4X4
C
Unique R2 (n=613, 11 sess)
unique R2
0.6
0.4
0.2
0
-0.2
0-1 s
1-3 s
init-dev
motion
rel 2-4Hz
abs δ
D
Effect of state on controller performance
Initial deviation
Motion
Rel 2-4 Hz
Abs δ
OL
CL
RMSE to ref (z)
Predictability ↓*
Regularizability ↓*
P ↓*
R n.s.
P n.s.
R ↓*
P ↓***
R ↓***
1
0.5
0
-0.5
Q1
Q2
Q3
Q4
sorted state quartile bins
E
Predictor kernel weights
+0.16
-0.16
stimulation site
F
Global predicts the counterfactual
Actual
Global (contra pred.)
Local
= A - G
ΔF/F (%)
2
0
-2
-4
R2te = 0.75
-1
0
1
2
3
time from stim (s)
G
CL rejects disturbances ***
rejection ratio (RR)
2
1
0.5
0.25
Mouse2a
Mouse1d
Mouse1c
Mouse2c
Mouse1a
Mouse2b
Mouse1b
Mouse1e
Mouse1g
Mouse3a
Mouse1f